Anova Test

Aim:

To perform a one-way ANOVA test in Python to determine whether there is a significant difference in the mean growth rates of plants under three different fertilizer treatments.

Procedure:

1. Import the required libraries — NumPy, SciPy, and statsmodels for statistical analysis.
2. Set a random seed to ensure reproducibility of the data generated.
3. Define the number of plants in each treatment group (e.g., 25 plants per treatment).
4. Generate random growth data for Treatment A, B, and C using normal distributions with different means and standard deviations.
5. Combine all growth data into a single array representing the complete dataset.
6. Assign treatment labels (A, B, and C) to each group of plant growth values.
7. Use the scipy.stats.f_oneway() function to perform a one-way ANOVA test on the three treatment groups.
8. Calculate and print the mean growth values for each treatment along with the F-statistic and P-value.
9. Compare the P-value with the significance level ($\alpha = 0.05$) to decide whether to reject or fail to reject the null hypothesis.
10. If the null hypothesis is rejected, perform a Tukey's HSD post-hoc test to identify which treatment means differ significantly.

Program:

```
import numpy as np

import scipy.stats as stats

from statsmodels.stats.multicomp import pairwise_tukeyhsd

np.random.seed(42)

n_plants = 25

growth_A = np.random.normal(loc=10, scale=2, size=n_plants)

growth_B = np.random.normal(loc=12, scale=3, size=n_plants)

growth_C = np.random.normal(loc=15, scale=2.5, size=n_plants)

all_data = np.concatenate([growth_A, growth_B, growth_C])

treatment_labels = ['A'] * n_plants + ['B'] * n_plants + ['C'] * n_plants

f_statistic, p_value = stats.f_oneway(growth_A, growth_B,
growth_C)

print("Treatment A Mean Growth:", np.mean(growth_A))

print("Treatment B Mean Growth:", np.mean(growth_B))

print("Treatment C Mean Growth:", np.mean(growth_C))

print()

print(f"F-Statistic: {f_statistic:.4f}")

print(f"P-Value: {p_value:.4f}")

alpha = 0.05

if p_value < alpha:

    print("Reject the null hypothesis: There is a significant difference
in mean growth rates among the three treatments.")
```

```
    tukey_results = pairwise_tukeyhsd(all_data, treatment_labels, alpha=0.05)
    print("\nTukey's HSD Post-hoc Test:")
    print(tukey_results)
else:
    print("Fail to reject the null hypothesis: There is no significant difference in mean growth rates among the three treatments.")
```

Output:

```
Treatment A Mean Growth: 9.672983882683818
Treatment B Mean Growth: 11.137680744437432
Treatment C Mean Growth: 15.265234904828972

F-Statistic: 36.1214
P-Value: 0.0000
Reject the null hypothesis: There is a significant difference in mean growth rates among the three treatments.

Tukey's HSD Post-hoc Test:
Multiple Comparison of Means - Tukey HSD, FWER=0.05
====================================================
group1 group2 meandiff p-adj   lower  upper  reject
----------------------------------------------------
     A      B   1.4647 0.0877 -0.1683 3.0977  False
     A      C   5.5923    0.0  3.9593 7.2252   True
```

```
    B     C   4.1276    0.0  2.4946 5.7605   True
----------------------------------------------------
```

Result:

Hence, a Python program was written to perform a one-way ANOVA test and post-hoc analysis, concluding that there is a **significant difference in the mean growth rates** of plants under the three fertilizer treatments.